*FinTech545: Quantitative Risk Management*
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# Modern Portfolio Optimization Approaches

## An analysis of Tradeoffs between Modern Portfolio Theory (Mean-Variance), Alternative Approaches (Risk-Parity), and Hybrid Models

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**ABSTRACT**

This paper compares mean-variance optimization (MVO) and risk parity (RP) through both a theoretical derivation and a set of back-tested algorithms for empirical evaluation. Furthermore, we also evaluate a hybrid Risk-Constrained MVO (RC-MVO) framework that delivers superior risk-adjusted performance under both frictionless and realistic execution conditions. All three strategies (RP, RP (ERC), and RC-MVO) are backtested on a seven-asset ETF universe over April 2008 – March 2026 across five distinct market regimes. MVO produces the highest unconstrained Sharpe ratio (0.843) but with extreme concentration and 13.5% monthly turnover, while ERC delivers stable, low-volatility performance that nonetheless suffers a structural failure during the 2022 rate-hiking regime as the equity-bond correlation turns positive. The proposed RC-MVO framework imposes a risk-budgeting constraint that caps each asset's risk contribution at $k/N$, retaining MVO's return-seeking direction while preventing concentration. With $k = 2$, RC-MVO achieves the highest Sharpe ratio of any strategy tested (0.922) and the smallest drawdown (19.33%), while reducing turnover by 30% relative to pure MVO. A live validation employing **QuantConnect API** is used through the paper to confirm our results under realistic execution.

All code, figures, and research, conducted both in Jupyter Notebook and in QuantConnect's backtesting environment is referenced in the bibliography.

**Key words.** Computational Finance – Operations Research – Optimization

## 1. Introduction

Financial markets offer a natural laboratory for quantitative research, combining methods from applied mathematics, statistics, and computer science to shape financial theory. A cornerstone of quantitative finance is **portfolio theory**—which refers to the study of how to allocate capital across financial assets in a mathematically sound and systematic way.

The field underwent a fundamental transformation in the mid-twentieth century. Prior to the 1950s, portfolio construction was largely an art, driven by arbitrary perceptions of asset quality. However, in 1952 Harry Markowitz published a paper on portfolio selection (Markowitz 1952), introducing Mean Variance Optimization, a rigorous mathematical framework that quantified both the return and risk of a portfolio in terms of its constituent assets' statistics. This framework, now known as Modern Portfolio Theory (MPT), demonstrated that diversification is not merely intuitive but mathematically optimal: and a key results demonstrated that portfolio's total risk is not the weighted average of individual asset risks, but rather a function of pairwise covariances. As the number of assets grows, covariance terms dominate, underscoring the central role of correlation in portfolio construction.

MPT remains one of the central frameworks of quantitative finance. Yet, along with the evolution of technology and financial theory, the underlying assumptions under Markowitz's MVO became increasingly challenged under academic and empirical scrutiny. Consider for example, asset returns exhibit well-documented behavior: volatility clustering, heavy tails, and time-varying correlations, all of which violate the multivariate normality that MPT assumes. More critically, MPT requires input parameters- one of which is an estimate of each asset's expected return, a parameter that is itself notoriously difficult to forecast with precision, and stochastic in nature. Small errors in these estimates are not merely absorbed by the optimizer—they are amplified, a phenomenon Michaud (1989) characterized as "error maximization." The result is portfolios that are theoretically optimal but practically unstable, often highly concentrated, and highly reliant on expectation estimate.

As such, these limitations motivated the development of alternative frameworks. Among them, Risk Parity (RP) emerged as a prominent counterpart to MPT. Rather than optimizing a risk-return tradeoff, Risk Parity constructs portfolios such that each asset contributes equally to total portfolio risk. It is important to note, however, that this is not merely a question of which approach performs better—rather, both frameworks reflect distinct investment philosophies. MPT assumes the investor has informed views on expected returns and seeks to exploit them; Risk Parity assumes the investor is agnostic about returns and instead seeks balanced risk exposure across economic regimes.

**Research objective.**
Through the process presented in this paper, we conduct a rigorous theoretical and empirical comparison of mean-variance optimization and risk parity portfolio construction methods.

On the theoretical side, we derive the analytical foundations of both frameworks—including the Lagrangian solution to the MVO quadratic program, the Capital Market Line, and the nonlinear Equal Risk Contribution (ERC) conditions underlying Risk Parity.

On the empirical side, we implement a rolling backtest of both portfolio manafgement approaches across an asset universe spanning U.S. equities, long-duration Treasuries, intermediate Treasuries, gold, and real estate investment trusts. We will then evaluate out-of-sample performance over different periods through 2005–2024. Windows will deliberately encompasses several distinct market regimes: the Global Financial Crisis of 2008, the COVID-19 shock of 2020, and the inflationary rate-hiking cycle of 2022—each of which stresses portfolio assumptions in different ways.

## 2. Mean-Variance Optimization

The foundation of modern portfolio theory rests on a deceptively simple premise: an investor cares only about the expected return and variance of their portfolio. Under this assumption, Markowitz (1952) formulated portfolio construction as a constrained optimization problem, giving rise to a mathematical framework that transformed the field.

### 2.1. Portfolio Return and Variance

Consider a universe of $N$ risky assets. Let $w = [w_1, w_2, \ldots, w_N]^\top \in \mathbb{R}^N$ denote the vector of portfolio weights, where $w_i$ represents the proportion of capital allocated to asset $i$. Let $\mu = [\mu_1, \mu_2, \ldots, \mu_N]^\top$ denote the vector of expected returns, and let $\Sigma \in \mathbb{R}^{N\times N}$ denote the covariance matrix of asset returns, with element $\sigma_{ij} = \mathrm{Cov}(r_i, r_j)$ and diagonal elements $\sigma_{ii} = \sigma_i^2$.

The expected return of the portfolio is the weighted average of individual asset expected returns. In summation notation:

$$\mu_p = \sum_{i=1}^{N} w_i \mu_i \tag{1}$$

which in matrix notation is expressed compactly as:

$$\mu_p = w^\top \mu \tag{2}$$

where $w$ is the $(N \times 1)$ column vector of portfolio weights and $\mu$ is the $(N \times 1)$ column vector of expected returns.

The true innovation of Markowitz lies in the formulation of portfolio variance. Unlike expected return, portfolio risk is not simply the weighted average of individual asset variances—it depends critically on how assets move together. In summation notation, the portfolio variance is:

$$\sigma_p^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{ij} \tag{3}$$

which in compact matrix form becomes:

$$\sigma_p^2 = w^\top \Sigma w \tag{4}$$

Observe, this formulation reveals a critical structural property of diversification. For a portfolio of $N$ assets, there are $N$ variance terms but $N^2 - N$ distinct covariance terms. As the number of assets increases, the portfolio's total risk becomes overwhelmingly dominated by the covariance structure, underscoring the paramount importance of correlation in portfolio construction. Portfolio standard deviation is then $\sigma_p = \sqrt{w^\top \Sigma w}$.

### 2.2. The Efficient Frontier

Since this research stemmed from research in financial economics, the next logical step was to frame the investor's problem through a utility function that trades off expected return against variance for a given level of risk aversion $\lambda > 0$: Markowitz (1952) framed the investor's problem through a utility function that trades off expected return against variance for a given level of risk aversion $\lambda > 0$:

$$\max_{w} \; U(w) = w^\top \mu - \frac{1}{2} \lambda \, w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1 \tag{5}$$

where $\mathbf{1}$ is an $N$-dimensional vector of ones and $\lambda$ is the risk aversion parameter. Solving this for varying levels of $\lambda$ traces out different points in risk-return space. The first-order conditions (KKT) of this problem are obtained by forming the Lagrangian:

$$\mathcal{L}(w, \gamma) = w^\top \mu - \frac{1}{2} \lambda \, w^\top \Sigma w - \gamma \left( \mathbf{1}^\top w - 1 \right) \tag{6}$$

Taking the derivative with respect to $w$ and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial w} = \mu - \lambda \Sigma w - \gamma \mathbf{1} = 0 \quad \Longrightarrow \quad w^* = \frac{1}{\lambda} \Sigma^{-1} (\mu - \gamma \mathbf{1}) \tag{7}$$

where $\gamma$ is pinned down by the constraint $\mathbf{1}^\top w^* = 1$. This closed-form solution confirms that for every value of $\lambda$, there exists a unique optimal portfolio. As $\lambda$ varies continuously, these solutions trace the efficient frontier. However, in practice, asking an investor to specify their $\lambda$ is unintuitive (Pazzula, 2026).

A more tractable reformulation therefore inverts the problem: rather than maximizing utility for a given risk tolerance, one fixes a target expected return $\mu^*$ and minimizes portfolio variance subject to achieving it. Note that the two formulations are equivalent—every solution to the utility problem for some $\lambda$ is also a solution to the variance-minimization problem for the corresponding $\mu^* = w^{*\top} \mu$. This duality yields the canonical constrained quadratic program:

$$\min_{w} \; \frac{1}{2} w^\top \Sigma w \quad \text{subject to} \quad w^\top \mu = \mu^*, \quad \mathbf{1}^\top w = 1 \tag{8}$$

This is a convex constrained optimization problem. In fact, it's a quadratic program (the objective function is quadratic) in $w$ and the constraints are linear, guaranteeing a unique global minimum for each target return $\mu^*$.

Analytically, one can solve this problem via Lagrange multipliers which yield a closed-form optimal weights expression as a function of $\mu^*$. By varying $\mu^*$ across a range of values and solving for the corresponding minimum-variance weights, one traces out what is known as the *efficient frontier*—the set of portfolios offering the maximum achievable expected return for each level of risk. When plotted in $(\sigma_p, \mu_p)$ space, this frontier is hyperbolic in shape, sometimes referred to as the "Markowitz bullet." Any portfolio on the upper portion of this curve is said to be *efficient*.

A fundamental result following from this derivation is the **Two-Fund Theorem**: any portfolio on the efficient frontier can be replicated as a convex combination of any two other distinct efficient portfolios. This means the entire frontier is spanned by just two funds, a result with deep implications for index investing and passive portfolio management.

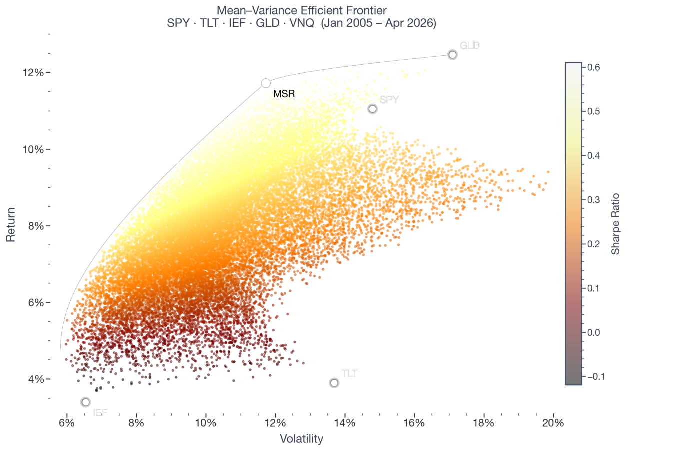



Fig. 1: **Efficient frontier for the five-asset universe.** The curve shows the minimum-variance frontier in $(\sigma_p, \mu_p)$ space for the portfolio of SPY, TLT, IEF, GLD, and VNQ, estimated using the full sample covariance matrix and historical mean returns (2005–2024). Then portfolios consisting solely of one asset are marked explicitly.

### 2.3. The Tangency Portfolio and Capital Market Line

The efficient frontier considers only risky assets. Introducing a risk-free asset with return $r_f$ and zero variance fundamentally changes the investment opportunity set. An investor may now combine any risky portfolio $p$ with the risk-free asset, forming convex combinations whose risk-return profile traces a straight line in $(\sigma_p, \mu_p)$ space, originating at $(0, r_f)$ and passing through $(\sigma_p, \mu_p)$.

**Sharpe Ratio.**
The slope of this line is the *Sharpe Ratio* of portfolio $p$:

$$SR_p = \frac{\mu_p - r_f}{\sigma_p} = \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \tag{9}$$

Investors prefer lines with the steepest slope, as these offer the greatest return per unit of risk. The optimal line, then, is referred to as the *Capital Market Line* (CML)—and its the one tangent to the efficient frontier of risky assets. The risky portfolio at the point of tangency is the ***Maximum Sharpe Ratio*** (MSR) portfolio, sometimes called the tangency portfolio or the super-efficient portfolio.

The optimization problem for the MSR portfolio is:

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \quad \text{subject to} \quad \mathbf{1}^\top w = 1, \quad w_i \geq 0 \tag{10}$$

This problem is non-convex due to the fractional objective. However, a change of variables transforms it into a tractable convex QP; such derivation is written below.

Define:

$$y = \frac{w}{w^\top \mu - r_f}, \qquad k = \frac{1}{w^\top \mu - r_f} \tag{11}$$

the problem becomes equivalent to:

$$\min_y \frac{1}{2} y^\top \Sigma y \quad \text{subject to} \quad (\mu - r_f \mathbf{1})^\top y = 1, \quad y_i \geq 0 \tag{12}$$

Once the optimal $y^*$ is found, the MSR weights are recovered by normalizing:

$$w^* = \frac{y^*}{\sum_i y_i^*} \tag{13}$$

Under the Capital Asset Pricing Model (CAPM) and the Efficient Market Hypothesis (EMH), the tangency portfolio coincides with the market portfolio—the value-weighted portfolio of all investable assets (Sharpe 1964). In practice, investors face borrowing constraints that limit their ability to reach points above the tangency portfolio on the CML, and most operate instead along the efficient frontier of risky assets.

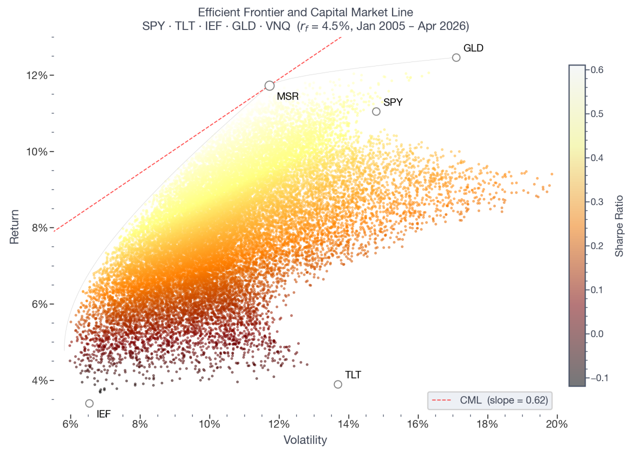



Fig. 2: **Efficient frontier, CML, and tangency portfolio.** The efficient frontier is shown alongside the Capital Market Line (red), constructed using a risk-free rate of $r_f = 4.3\%$ (approximate 2026 10Y Treasury Note). The tangency portfolio (MSR) is marked.

### 2.4. Estimation Error and Practical Limitations

Despite its theoretical depth, MVO suffers from a fundamental practical vulnerability. The optimizer requires two sets of **inputs**: the vector of **expected returns** $\mu$ and the **covariance matrix** $\Sigma$. Of these, expected returns are notoriously difficult to estimate with precision—empirically, return volatility dwarfs the mean return over short horizons, making $\mu$ estimates highly noisy.

Consider, forecasting returns is not a 'solved problem,' and although there are multiple approaches to deriving the expected return for an asset, these are themselves uncertain and intrinsically stochastic.

Critically, the MVO optimizer does not merely absorb estimation error, but because its objective is to find the *maximum* of the Sharpe Ratio surface, it aggressively allocates capital toward assets with even slightly overestimated expected returns and away from those with slightly underestimated returns. The solver will actively seeks out and amplifies estimation errors in the inputs. Michaud (1989) termed this phenomenon *error maximization*: MVO is theoretically optimal under the impossible condition that its inputs are known with perfect certainty.

As DeMiguel et al. (2009) showed in a comprehensive study, naive $1/N$ equal-weight portfolios outperform MVO on a risk-adjusted basis across a wide range of datasets and estimation windows—a sobering result for a framework that is supposed to be "optimal." These limitations motivate the search for more robust allocation frameworks.

Second moments of return distributions—variances and covariances—are empirically far more stable and predictable than first moments. A natural question arises: **can one construct a principled portfolio using only the covariance matrix, dispensing with expected returns entirely**?

This is precisely the motivation for Risk Parity, which we'll develop in Section 3.

Figure 3 illustrates the error maximization problem directly. We held the covariance matrix fixed, and perturbed SPY's expected return by ±2% and then recomputed the MSR portfolio. Despite this being a modest perturbation, the optimal weights shift dramatically across all five assets. The equal-weight portfolio, which uses no return estimates, is entirely unaffected.

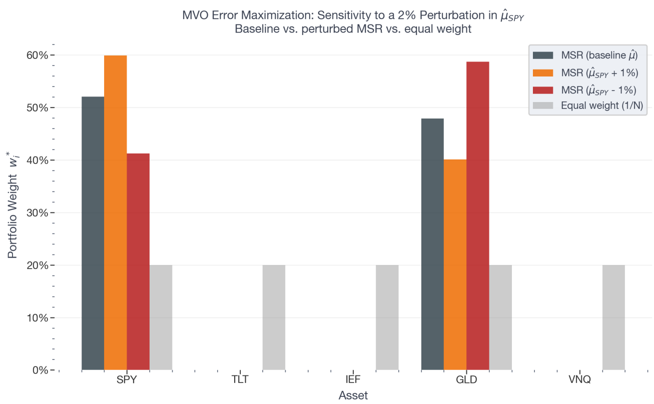



Fig. 3: **MVO sensitivity to a 2% perturbation in $\hat{\mu}_{\mathrm{SPY}}$.** A modest change in a single return estimate reshuffles the entire portfolio, illustrating Michaud's error maximization phenomenon (Michaud 1989).

## 3. Risk Parity

The previous section established that mean-variance optimization, despite its theoretical elegance, suffers from a fundamental practical vulnerability: it is acutely sensitive to expected return estimates, which are notoriously difficult to forecast. This observation motivates a different question entirely. Rather than asking *how do we optimally trade off risk and return*, Risk Parity asks: *if we cannot reliably estimate returns, can we at least ensure that our portfolio is balanced in terms of risk*? The answer to this question leads to a framework that is both mathematically rigorous and intuitively compelling.

### 3.1. Intuition: From Capital Diversification to Risk Diversification

To understand why Risk Parity emerged, consider a simple and extremely common portfolio: the classic 60/40 allocation, where 60% of capital is invested in equities and 40% in bonds. On the surface, this appears diversified—after all, capital is split across two asset classes. But consider what happens when we examine the *risk* split rather than the capital split.

Equities historically exhibit annualized volatility of approximately 15–18%, while investment-grade bonds exhibit volatility closer to 5–8%. Because equities are so much more volatile, they contribute the overwhelming majority of the portfolio's total risk—often upwards of 80–90%—despite receiving only 60% of the capital. The 40% bond allocation, while meaningful in dollar terms, is nearly irrelevant in risk terms. In effect, the 60/40 portfolio is not a balanced portfolio at all; it is an equity portfolio with a small fixed income overlay.

Risk Parity challenges this conventional wisdom by redefining what "diversification" means. Rather than allocating capital equally—or according to return forecasts—Risk Parity allocates capital such that each asset contributes *equally to total portfolio risk*. The insight is simple but powerful: true diversification is diversification of risk, not of dollars (Qian 2005). This philosophy was formalized and popularized by Ray Dalio at Bridgewater Associates through the All Weather portfolio, designed to perform robustly across all macroeconomic regimes by balancing risk contributions from assets that respond differently to growth and inflation shocks (Dalio 2004).

### 3.2. Risk Decomposition: The Mathematics of Risk Contribution

To implement Risk Parity, one must first be able to decompose total portfolio risk into contributions from each individual asset. This requires a precise mathematical definition of what it means for an asset to "contribute" to portfolio risk.

Recall that portfolio volatility is given by:

$$\sigma_p(w) = \sqrt{w^\top \Sigma w} \tag{14}$$

This is a scalar function of the weight vector $w$. A natural question is: how does $\sigma_p$ change if we marginally increase the weight of asset $i$? The answer is given by the partial derivative of $\sigma_p$ with respect to $w_i$, which defines the *Marginal Contribution to Risk* (MCR) of asset $i$:

$$\mathrm{MCR}_i = \frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sqrt{w^\top \Sigma w}} = \frac{\sum_{j=1}^{N} w_j \sigma_{ij}}{\sigma_p} \tag{15}$$

where $(\Sigma w)_i$ denotes the $i$-th element of the vector product $\Sigma w$. Intuitively, $\mathrm{MCR}_i$ measures how much total portfolio volatility increases for an infinitesimally small increase in the weight of asset $i$, holding all other weights fixed. It captures not just asset $i$'s own volatility but also how it co-moves with the rest of the portfolio—an asset that is highly correlated with the existing portfolio will have a large MCR.

The *Total Risk Contribution* (RC) of asset $i$ is then defined as its weight multiplied by its marginal contribution:

$$\mathrm{RC}_i = w_i \cdot \mathrm{MCR}_i = \frac{w_i \cdot (\Sigma w)_i}{\sigma_p} \tag{16}$$

This definition has a beautiful and non-obvious property: the individual risk contributions sum exactly to total portfolio volatility. This follows directly from **Euler's Homogeneous Function Theorem**. Since $\sigma_p(w)$ is a homogeneous function of degree one in $w$—meaning $\sigma_p(\lambda w) = \lambda\sigma_p(w)$ for any scalar $\lambda > 0$— Euler's theorem guarantees:

$$\sigma_p(w) = \sum_{i=1}^{N} w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^{N} \mathrm{RC}_i \tag{17}$$

This is a remarkable result. It provides an exact, additive decomposition of a non-additive risk measure—portfolio volatility is not the sum of individual volatilities, but it *is* the sum of individual risk contributions. In vector notation, defining the component standard deviation vector as:

$$\mathrm{CSD} = w \odot \frac{\Sigma w}{\sqrt{w^\top \Sigma w}} \tag{18}$$

where $\odot$ denotes element-wise multiplication, we have $\mathbf{1}^\top\mathrm{CSD} = \sigma_p$ (Roncalli 2013).

### 3.3. The Equal Risk Contribution Portfolio

Armed with a precise definition of risk contribution, the Risk Parity objective can be stated formally. The *Equal Risk Contribution* (ERC) portfolio is the weight vector $w^*$ such that every asset contributes identically to total portfolio risk:

$$\mathrm{RC}_i(w) = \mathrm{RC}_j(w) \quad \forall\, i, j \in \{1, \ldots, N\} \tag{19}$$

Since the risk contributions sum to $\sigma_p$, this is equivalent to requiring each asset to contribute exactly $1/N$ of total risk:

$$\mathrm{RC}_i(w) = \frac{\sigma_p}{N} \quad \forall\, i \tag{20}$$

This is an appealingly democratic condition: every asset has an equal "vote" in determining portfolio risk, regardless of its size or return characteristics.

Unlike the MVO problem, which admits a closed-form solution via Lagrange multipliers, the ERC conditions form a *nonlinear system of equations* in $w$. There is no analytical solution in general. The problem is therefore formulated as a numerical optimization, minimizing the sum of squared deviations between all pairwise risk contributions:

$$\min_w \sum_{i=1}^{N} \left(w_i(\Sigma w)_i - w_j(\Sigma w)_j\right)^2 \quad \forall\, i, j \tag{21}$$

subject to $\mathbf{1}^\top w = 1$ and $w_i \geq 0$. An equivalent and often more numerically stable formulation minimizes the sum of squared deviations between each asset's fractional risk contribution and its target budget $b_i = 1/N$:

$$\min_w \sum_{i=1}^{N} \left(\frac{w_i(\Sigma w)_i}{w^\top \Sigma w} - b_i\right)^2 \tag{22}$$

Both formulations are nonlinear and non-convex, requiring iterative numerical solvers such as Sequential Least Squares Programming (SLSQP). A useful warm-start for the optimizer is the *inverse volatility portfolio*, which sets $w_i \propto 1/\sigma_i$. This approximates the ERC solution when all pairwise correlations are equal, and provides initial weights close enough to the optimum to ensure reliable convergence (Roncalli 2013).

Figure 4 illustrates the core distinction between the three portfolio construction approaches. The equal-weight portfolio distributes capital uniformly but produces highly unequal risk contributions, with high-volatility assets dominating. The MSR portfolio concentrates both capital and risk in two assets. The ERC portfolio, by construction, achieves equal risk contributions across all five assets.

### 3.4. Implicit Assumptions and Limitations

Risk Parity is often presented as relatively less assumption dependent because it does not require expected return estimates.

This characterization is, however, not absolute. Consider, by constructing a portfolio that equalizes risk contributions, Risk Parity implicitly embeds a specific and strong view about returns: that all assets offer the same long-run risk-adjusted return, i.e., the same Sharpe ratio. It is also important to note that the Covariance Matrix itself is an estimate, and several reconstruction techniques as well as exponential weights could lead to different numerical etimates– albeit is significantly more stable thant expectation of returns.

To see why, note that if all assets had equal Sharpe ratios, then the portfolio that maximizes the overall Sharpe ratio would be precisely the ERC portfolio (Qian 2005). Risk Parity does not eliminate return assumptions — it replaces the granular, difficult task of forecasting individual returns with a single, broad philosophical belief: *markets efficiently price risk in the long run.*

A second practical consideration is leverage. Because Risk Parity assigns large weights to low-volatility assets such as government bonds, the resulting portfolio tends to have lower expected return than a traditional 60/40 allocation. Institutional implementations of Risk Parity, such as Bridgewater's All Weather fund, therefore apply leverage to scale the portfolio's risk and return up to a target level. Leverage introduces its own risks—path dependency, margin calls during drawdowns, and amplified losses when cross-asset correlations spike toward unity during systemic crises, precisely the environments where diversification is most needed.

A third consideration, as previously hinted, is the stability of the covariance matrix. The equity-bond correlation, a cornerstone of risk parity diversification, was reliably negative throughout the 2000–2021 period but turned sharply positive in 2022 as the Federal Reserve raised rates aggressively, causing both equities and long-duration bonds to fall simultaneously. Our empirical analysis in Section 5 directly examines this regime shift and its implications for Risk Parity performance.

Heuristically, as a slight detour, one might find that returns of bonds and equities at the time this paper is written are not all that de-correlated either. One phenomenon driving SPY returns is the boom or Artificial Intelligence, which is precisely a major component driving debt as well. Therefore, it would not be too far to consider that both equities and debt are, at this moment, highly concentrated in technology companies or other

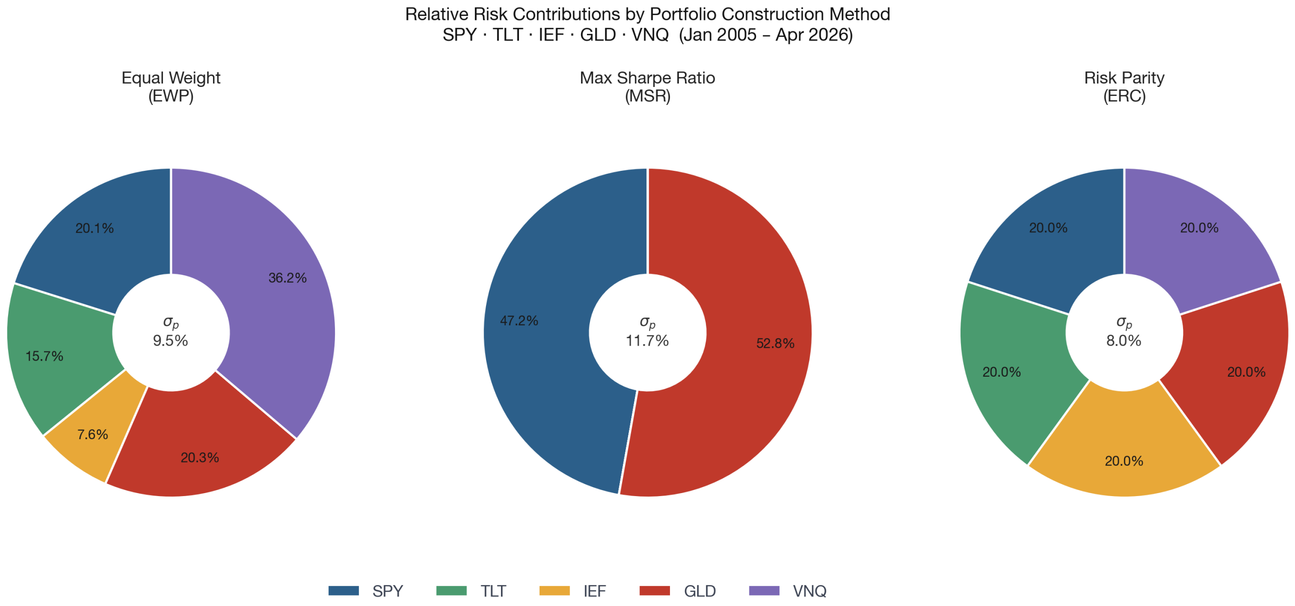



Fig. 4: **Relative risk contributions by portfolio construction method.** Each donut shows the fraction of total portfolio volatility attributable to each asset under the equal-weight (EWP), maximum Sharpe ratio (MSR), and equal risk contribution (ERC) portfolios, estimated using the full sample covariance matrix (Jan 2005 – Apr 2026). Portfolio volatility $\sigma_p$ is annotated at the center of each donut. The ERC portfolio achieves equal risk slices by construction; the others do not.

institutions borrowing for AI-related spending.

A natural extension of the volatility-based Risk Parity framework is to replace $\sigma_p$ with a more robust risk measure. Since asset returns exhibit heavy tails and asymmetry—properties that variance fails to capture—one can redefine risk contributions using *Expected Shortfall* (ES), also known as Conditional Value-at-Risk (CVaR). ES is a coherent risk measure and, unlike VaR, produces a convex optimization surface amenable to numerical minimization. The marginal ES contribution of asset $i$ can be estimated via finite differences:

$$\frac{\partial \mathrm{ES}}{\partial w_i} \approx \frac{\mathrm{ES}(w_1, \ldots, w_i + \varepsilon, \ldots, w_N \mid R) - \mathrm{ES}(w \mid R)}{\varepsilon} \tag{23}$$

where $R$ denotes a matrix of simulated or historical returns. The component ES contributions $\mathrm{CES}_i = w_i \cdot \partial \mathrm{ES}/\partial w_i$ then replace the CSD vector in the SSE objective, and the optimization proceeds identically (Roncalli 2013).

### 3.5. Risk Parity as a Generalization: Risk Budgeting

The ERC framework generalizes naturally to the case where assets are not required to contribute equally but rather according to a pre-specified *risk budget* vector $b = [b_1, b_2, \ldots, b_N]^\top$ with $\sum_i b_i = 1$. The risk budgeting problem then seeks weights such that:

$$\frac{\mathrm{RC}_i(w)}{\sigma_p} = b_i \quad \forall\, i \tag{24}$$

ERC is the special case $b_i = 1/N$ for all $i$. This generalization is practically important: a portfolio manager who has conviction that equities will outperform in the medium term can express this view by assigning equities a larger risk budget, without abandoning the risk-based framework entirely. The optimization proceeds identically, replacing the equal target $b_i = 1/N$ with the manager's chosen budget vector.

This framework thus occupies an interesting middle ground between MVO and pure Risk Parity.

It is more robust than MVO because it does not require precise return forecasts, but more flexible than ERC because it accommodates investor views through the risk budget. In our empirical analysis, we focus on the pure ERC case ($b_i = 1/N$) as the canonical Risk Parity implementation, leaving risk budgeting extensions to Section 7.

The numerical implementation of the general risk budgeting problem proceeds by modifying the SSE objective to account for the unequal target budgets. Define the budget-adjusted Component Standard Deviation vector:

$$\mathrm{CSD}_i^* = \frac{\mathrm{CSD}_i}{b_i} = \frac{w_i(\Sigma w)_i}{b_i\, \sigma_p} \tag{25}$$

The risk budgeting optimization then minimizes the sum of squared deviations of this adjusted vector around its mean $\overline{\mathrm{CSD}^*}$:

$$\min_w \sum_{i=1}^{N} \left(\mathrm{CSD}_i^* - \overline{\mathrm{CSD}^*}\right)^2 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0 \tag{26}$$

When $b_i = 1/N$ for all $i$, this reduces exactly to the ERC formulation in Eq. (8), confirming that ERC is a special case of the general risk budgeting framework. The optimization is solved iteratively via SLSQP, initialized at the inverse volatility weights $w_i \propto 1/\sigma_i$.

## 4. Data and Methodology

This section describes the asset universe, data sources, sample period, parameter estimation methodology, and the programming workflow used to compare mean-variance optimization and

risk parity empirically.
The pipeline is organized across two environments: a Python/Jupyter environment for static analysis, and the QuantConnect Backtesting environments for the rolling backtest and live strategy simulation.

### 4.1. Asset Universe

The empirical analysis is conducted on a universe of seven exchange-traded funds (ETFs) spanning five distinct asset classes. The universe is reported in Table 1.

Table 1: Asset universe and annualized statistics, Feb 2005–Mar 2026. Sharpe ratios use average 3-month T-bill rate of 1.71%.

| Ticker | Description | Ret. | Vol. | SR |
|---|---|---|---|---|
| SPY | SPDR S&P 500 ETF | 11.05% | 14.79% | 0.63 |
| QQQ | Invesco Nasdaq-100 ETF | 15.40% | 18.09% | 0.76 |
| TLT | iShares 20+ Yr Treasury | 3.90% | 13.69% | 0.16 |
| IEF | iShares 7–10 Yr Treasury | 3.39% | 6.53% | 0.26 |
| TIP | iShares TIPS Bond ETF | 3.44% | 5.67% | 0.30 |
| GLD | SPDR Gold Shares | 12.46% | 17.10% | 0.63 |
| VNQ | Vanguard Real Estate ETF | 9.26% | 21.58% | 0.35 |

This universe spans meaningfully distinct risk factors. SPY and QQQ provide U.S. equity beta at different levels of concentration; QQQ is more technology-weighted with historically higher return and volatility. TLT and IEF provide duration risk at different maturities. TIP provides inflation-linked exposure with significantly lower interest rate sensitivity than TLT, offering resilience to rate-hiking regimes. GLD provides commodity and inflation-hedge exposure with historically low correlation to both equities and bonds. VNQ provides real estate exposure with distinct yield and rate characteristics.

The spread of annualized volatilities—from 6.5% (IEF) to 21.6% (VNQ)—is precisely what induces divergent behavior between MVO and Risk Parity: MVO will aggressively exploit return differences while ERC will systematically overweight low-volatility assets.

### 4.2. Data Sources and Returns

Monthly adjusted closing prices are sourced from Yahoo Finance via `yfinance` for the Jupyter environment, and from QuantConnect's proprietary data library via `QuantBook` for the rolling backtest. Monthly *simple* returns are computed as:

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} \tag{27}$$

Simple returns are used rather than log returns because portfolio aggregation requires linearity: $r_{p,t} = \sum_i w_i r_{i,t}$ holds exactly for simple returns but not for log returns, since $\log(1 + r_p) \neq \sum_i w_i \log(1 + r_i)$ in general. The Markowitz framework's moment conditions $\mu_p = w^\top \mu$ and $\sigma_p^2 = w^\top \Sigma w$ are therefore only valid under simple returns (Meucci 2010).

The sample spans February 2005 through March 2026, yielding 254 monthly observations. This window encompasses qualitatively distinct market regimes: the Global Financial Crisis (2008–2009), the post-crisis zero interest rate environment (2009–2015), the COVID-19 crash and recovery (2020), and the inflationary rate-hiking cycle of 2022—each stressing portfolio assumptions in different ways.

### 4.3. Parameter Estimation

Both MVO and Risk Parity require estimates of the covariance matrix $\Sigma$; MVO additionally requires the expected return vector $\mu$. The quality of these estimates is the central practical challenge of portfolio construction.

**Expected returns.** We use the sample mean of trailing 36-month returns as the expected return estimate:

$$\hat{\mu}_t = \frac{1}{36} \sum_{\tau=t-35}^{t} r_\tau \tag{28}$$

This is the simplest and most transparent estimator, and the one used as a baseline in the academic portfolio optimization literature (DeMiguel et al. 2009; Roncalli 2013). It is well-known that sample mean estimates are extremely noisy — Chopra and Ziemba (1993) showed that errors in means are roughly ten times more costly than errors in variances in their impact on portfolio performance. More sophisticated alternatives include CAPM-implied returns, Black-Litterman estimates, and shrinkage toward a prior. We use the sample mean as our baseline and discuss robustness in Section 7.
**Covariance matrix.** We use the sample covariance matrix of trailing 36-month returns:

$$\hat{\Sigma}_t = \frac{1}{35} \sum_{\tau=t-35}^{t} (r_\tau - \hat{\mu}_t)(r_\tau - \hat{\mu}_t)^\top \tag{29}$$

The 36-month window provides 36 observations for a 7-asset covariance matrix, yielding a positive definite estimate with reasonable stability. This is consistent with industry practice for monthly-rebalancing multi-asset strategies. Two well-known alternatives are worth noting. The *EWMA* estimator (J.P. Morgan/Reuters 1996) assigns exponentially decaying weights to past observations:

$$\hat{\Sigma}_t^{EWMA} = \lambda \hat{\Sigma}_{t-1}^{EWMA} + (1 - \lambda) r_{t-1} r_{t-1}^\top, \quad \lambda \approx 0.97 \tag{30}$$

with higher $\lambda$ producing slower decay. EWMA is more responsive to recent volatility regimes but discards useful long-run information. The *Ledoit-Wolf shrinkage* estimator (Ledoit and Wolf 2004) addresses the well-known instability of sample covariance matrices by shrinking toward a structured target:

$$\hat{\Sigma}_t^{LW} = \alpha \hat{\Sigma}_t^{sample} + (1 - \alpha) \hat{\Sigma}^{target} \tag{31}$$

where $\alpha \in [0, 1]$ is chosen optimally to minimize mean-squared estimation error. Shrinkage provably reduces estimation error and is widely used in practice, particularly for larger asset universes. For our 7-asset universe with 36 monthly observations, sample covariance is sufficiently well-conditioned that shrinkage provides limited additional benefit.

### 4.4. Computational Workflow

The empirical pipeline spans three computational environments as illustrated in Figure 5.

### 4.5. Rolling Walk-Forward Backtest

Both portfolios are evaluated under a rolling walk-forward framework producing genuine out-of-sample performance estimates. The first rebalancing date is March 2008, leaving

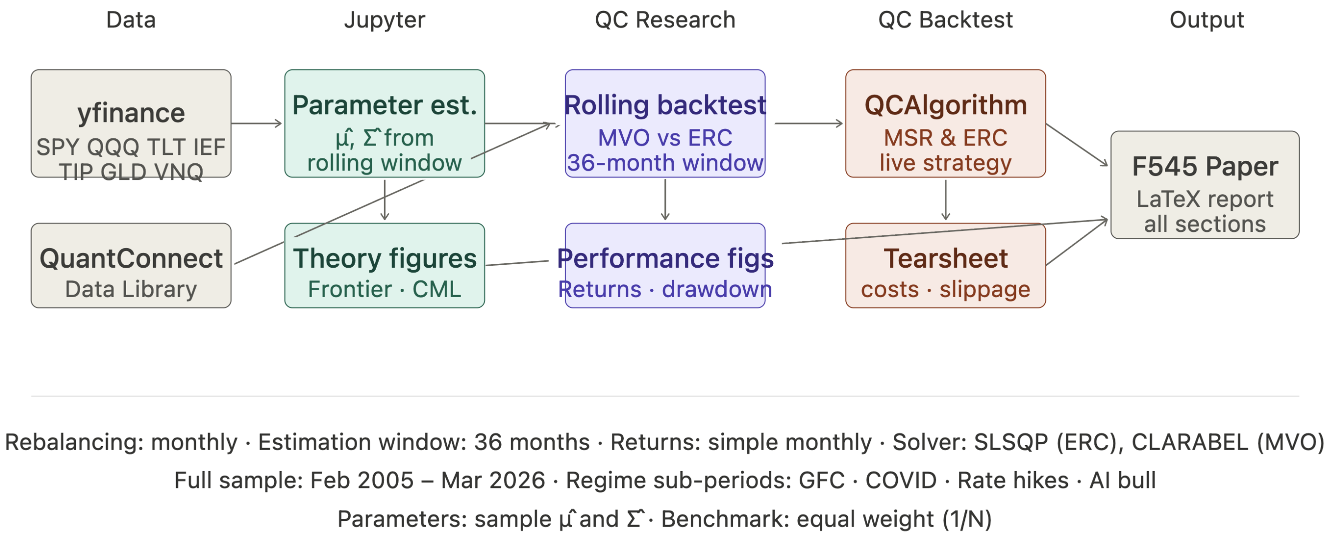



Fig. 5: **Computational workflow.** Static theoretical analysis and figures are produced in Python/Jupyter using `yfinance` data. The rolling walk-forward backtest and performance figures are implemented in the QuantConnect Research environment. The live strategy simulation with transaction costs and slippage is implemented as a `QCAlgorithm` environment.

36 months for initial estimation. The out-of-sample period runs April 2008 through March 2026, yielding 216 monthly observations.

The **MVO portfolio** maximizes the Sharpe ratio at each step via convex QP (Section 2.3), solved using Python's `cvxpy` library with `CLARABEL` under long-only constraints ($w \geq 0$, $\mathbf{1}^\top w = 1$). The contemporaneous 3-month T-bill rate is used as $r_f$.

The **ERC portfolio** solves the risk budgeting optimization (Section 3.3) at each step using `scipy.optimize` SLSQP, initialized at inverse volatility weights. No expected return input is required.

In addition to the full-sample backtest, we conduct independent sub-period analyses across five distinct market regimes: the Global Financial Crisis (Jan 2007–Dec 2009), the post-GFC bull market (Jan 2010–Dec 2013), the COVID-19 crash and recovery (Jan 2020–Dec 2021), the inflationary rate-hiking cycle (Jan 2022–Dec 2023), and the AI-driven bull market (Jan 2024–Mar 2026). The methodology remains identical across all periods; only the start and end dates change.

### 4.6. Performance Metrics

Performance is evaluated on the out-of-sample return series using:

**Annualized return and volatility:**

$$\bar{r}^p = 12 \cdot \frac{1}{T}\sum r_t^p, \qquad \hat{\sigma}^p = \sqrt{12} \cdot \sqrt{\frac{1}{T-1}\sum (r_t^p - \bar{r}^p)^2} \quad (32)$$

**Sharpe ratio, maximum drawdown, and Calmar ratio:**

$$SR^p = \frac{\bar{r}^p - \bar{r}_f}{\hat{\sigma}^p}, \quad MDD^p = \max_t\left(\max_{\tau \leq t} X_\tau - X_t\right), \quad \mathrm{Cal}^p = \frac{\bar{r}^p}{MDD^p} \quad (33)$$

where $X_t = \prod_\tau (1 + r_\tau^p)$ is the cumulative wealth index. Rolling 12-month Sharpe ratios are examined to assess regime-dependent performance stability.

**Monthly turnover:**

$$\mathrm{TO}_t = \frac{1}{2}\sum_{i=1}^{N} |w_{i,t} - w_{i,t-1}^+| \quad (34)$$

where $w_{i,t-1}^+$ denotes the weight of asset $i$ at the end of month $t-1$ after price drift but before rebalancing. Average monthly turnover quantifies implicit transaction costs.

## 5. Empirical Results

This section presents the empirical performance of the rolling walk-forward backtest over the full out-of-sample period April 2008 through March 2026. We compare three portfolio construction methods: maximum Sharpe ratio mean-variance optimization (MVO), equal risk contribution risk parity (ERC), and the equal-weight benchmark (EW). All portfolios are rebalanced monthly using a 36-month rolling estimation window, under long-only constraints, with no transaction costs assumed in this section.

### 5.1. Full-Sample Performance

Table 2: Full-sample out-of-sample performance, April 2008 – March 2026.

| Metric | MVO | ERC | EW |
|---|---|---|---|
| Ann. Return | 10.29% | 7.04% | 8.10% |
| Ann. Volatility | 10.71% | 7.75% | 9.51% |
| Sharpe Ratio | 0.843 | 0.737 | 0.712 |
| Maximum Drawdown | 23.85% | 19.53% | 22.15% |
| Calmar Ratio | 0.431 | 0.360 | 0.366 |
| Avg Monthly Turnover | 13.47% | 1.78% | 0.00% |

The aggregate results exhibit the trade-offs predicted by theory. MVO achieves the highest annualized return (10.29%) and

Sharpe ratio (0.843), reflecting its ability to concentrate capital in high-Sharpe assets. ERC delivers a more muted return (7.04%) but with substantially lower volatility (7.75%) and the smallest maximum drawdown (19.53%). The equal-weight benchmark performs in between on return but produces the lowest Sharpe ratio (0.712), consistent with the theoretical prediction that 1/N ignores both return and risk information.

The most striking result is the disparity in *turnover*: MVO turns over 13.5% of the portfolio each month — approximately 162% annually — while ERC turns over only 1.8% monthly (roughly 21% annually), a factor of 7.5× difference. This finding has significant practical implications. At a conservative estimate of 5–10 basis points round-trip transaction cost per ETF trade, MVO's advantage in raw Sharpe ratio would shrink substantially in a realistic implementation. We revisit this in Section 7 when we deploy our startegy through QC API with realistic transaction costs.

### 5.2. Cumulative Performance and Drawdowns

Figure 6 plots the cumulative wealth index of each strategy assuming \$1 invested at inception. By March 2026, the MVO portfolio had grown to \$5.59, the ERC portfolio to \$3.36, and the equal-weight portfolio to \$3.97.

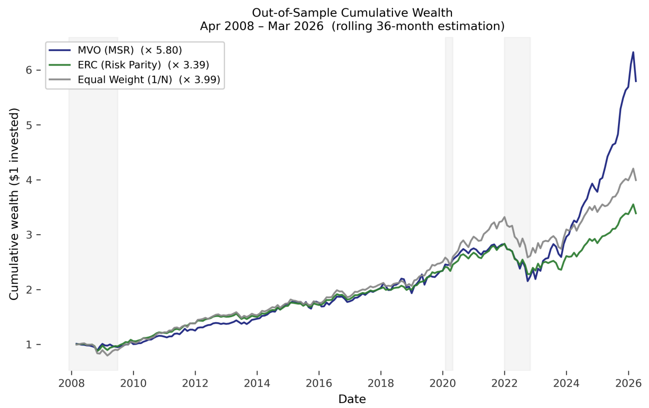



Fig. 6: **Out-of-sample cumulative wealth.** Cumulative wealth index (\$1 invested Apr 2008) for MVO, ERC, and equal-weight portfolios. Shaded regions indicate recessionary or crisis periods (GFC, COVID-19, 2022 rate shock).

Figure 7 plots the drawdown trajectory of each strategy. ERC achieves the smallest maximum drawdown (19.53%) and recovers fastest in most crisis periods, validating its design as a risk-balanced strategy. MVO's deepest drawdown occurs in 2022 when both its concentrated equity (QQQ) and gold (GLD) positions declined simultaneously — a regime in which MVO's diversification assumption broke down.

### 5.3. Rolling Sharpe Ratio

Aggregate Sharpe ratios obscure substantial regime variation. Figure 8 plots the rolling 12-month Sharpe ratio computed as $SR_t = \overline{r - r_f}/\sigma_r \cdot \sqrt{12}$ over a trailing 12-month window, following the standard convention used in industry backtesting tools (QuantStart Team 2017). ERC produces the most stable rolling Sharpe trajectory, consistent with its construction as a low-turnover, risk-balanced strategy. MVO exhibits dramatic swings,

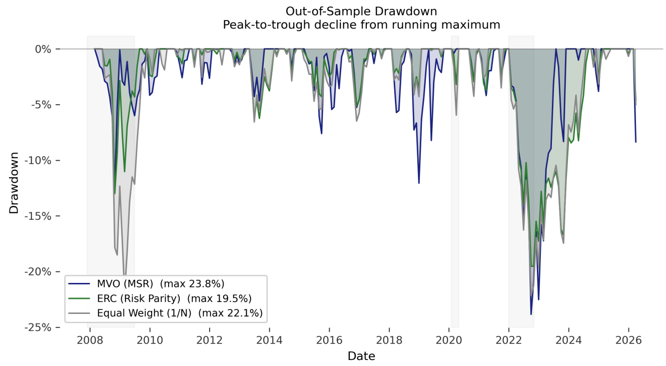



Fig. 7: **Out-of-sample drawdown.** Peak-to-trough decline from running maximum, in percent.

including peaks above 2.5 during the AI-driven equity rally and troughs near −2 during the 2022 rate shock. This volatility in the Sharpe ratio itself — not merely in returns — is itself a form of estimation risk, reflecting the instability of the underlying weight allocations.

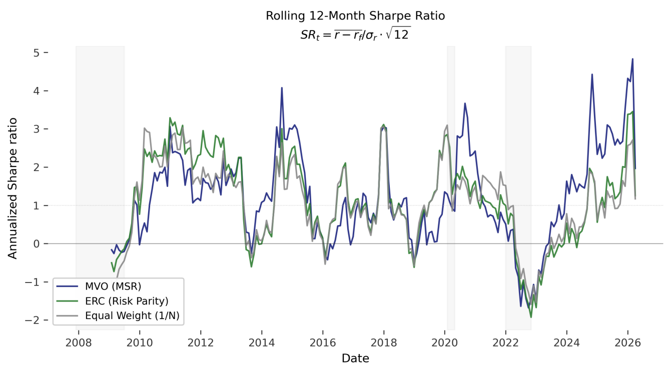



Fig. 8: **Rolling 12-month Sharpe ratio.** $SR_t$ computed over the trailing 12 months of monthly returns, annualized by $\sqrt{12}$. ERC produces the most stable Sharpe trajectory; MVO is the most volatile.

### 5.4. Regime-Dependent Performance

Aggregate metrics smooth across qualitatively different regimes. Table 3 reports performance over five distinct sub-periods: the Global Financial Crisis (GFC, 2007–2009), the post-GFC bull market (2010–2013), the COVID-19 shock and recovery (2020–2021), the inflationary rate-hiking cycle (2022–2023), and the AI-driven equity rally (2024–2026).

The regime decomposition reveals a fundamental asymmetry. ERC outperforms MVO during the GFC (3.72% vs 0.81% return, 0.256 vs 0.017 Sharpe), demonstrating risk parity's defensive properties when equity-bond diversification works as expected.

The rate-hiking regime of 2022–2023 is the most revealing. ERC *loses* 3.40% annually with a Sharpe of −0.574, while MVO eked out a modest 3.96% gain. This inversion of the typical pattern reflects the breakdown of the equity-bond correlation structure: as the Federal Reserve raised rates aggressively, both equities and long-duration Treasuries fell simultaneously.

Table 3: Regime-dependent performance. Sharpe ratios use the average 3-month T-bill rate specific to each sub-period.

| Period | Strategy | Return | Volatility | Sharpe | Max DD |
|---|---|---|---|---|---|
| GFC (2007–2009) | MVO | 0.81% | 10.92% | 0.017 | 12.54% |
| | ERC | 3.72% | 12.08% | 0.256 | 13.00% |
| | EW | 3.20% | 15.25% | 0.169 | 21.96% |
| Post-GFC (2010–2013) | MVO | 9.69% | 5.92% | 1.624 | 4.67% |
| | ERC | 8.89% | 5.35% | 1.646 | 6.24% |
| | EW | 9.97% | 6.91% | 1.432 | 6.56% |
| COVID (2020–2021) | MVO | 9.68% | 6.23% | 1.526 | 4.19% |
| | ERC | 9.79% | 6.94% | 1.387 | 3.75% |
| | EW | 14.67% | 9.52% | 1.524 | 5.95% |
| Rate Hikes (2022–2023) | MVO | 3.96% | 18.17% | 0.021 | 21.25% |
| | ERC | −3.40% | 12.18% | −0.574 | 16.60% |
| | EW | −2.46% | 14.58% | −0.415 | 18.39% |
| AI Bull (2024–2026) | MVO | 30.89% | 11.44% | 2.318 | 8.35% |
| | ERC | 11.91% | 6.70% | 1.125 | 4.59% |
| | EW | 11.67% | 7.56% | 0.966 | 5.01% |

Finally, the AI-driven bull market of 2024–2026 produced MVO's best regime: a 30.89% annualized return with a Sharpe of 2.318. By this period, the rolling 36-month window had captured QQQ's exceptional performance, leading the optimizer to allocate aggressively to it. ERC, by construction unable to overweight any single asset significantly, captured only 11.91%.

Figure 9 visualizes this distinction directly. The MVO panel shows abrupt rotations — entire allocations to QQQ, GLD, or TIP appearing and disappearing across rebalancing dates. The ERC panel shows a slowly evolving allocation in which all seven assets maintain non-trivial weights throughout the sample, with TIP and IEF receiving the largest average allocations (22.3% and 24.8% respectively) due to their low volatility.

### 5.5. Summary of Empirical Findings

The empirical results can be summarized in five points. First, MVO produced the highest aggregate Sharpe ratio (0.843) but at the cost of extreme concentration (HHI $\approx$ 0.52) and high turnover (13.5% monthly). Second, ERC produced the lowest volatility and shallowest drawdowns across the full sample, validating its design as a risk-balanced strategy. Third, regime analysis reveals that MVO's edge is concentrated in periods where its overweighted assets perform exceptionally (the AI bull market), while ERC outperforms during traditional risk-off periods (the GFC). Fourth, the 2022 rate-hiking regime exposed a structural vulnerability of risk parity: the breakdown of the equity-bond correlation eliminated its diversification benefit and produced negative absolute returns. Fifth, the equal-weight benchmark, despite using no return or risk information, performed competitively in several regimes — consistent with DeMiguel et al. (2009). We discuss the implications of these findings in Section 7.

## 6. Extensions: Risk-Constrained MVO and Transaction Costs

The empirical results in Section 5 reveal a fundamental trade-off: MVO produces the highest raw Sharpe ratio but at the cost of extreme concentration, while ERC provides stability at the cost of foregone return.

This section explores two extensions designed to address these limitations: a hybrid *risk-constrained MVO* formulation that imposes a risk-parity-inspired cap on per-asset risk contribution (inequality constraint), and a transaction-cost-adjusted analysis that quantifies how each strategy's performance changes under realistic implementation costs.

### 6.1. Risk-Constrained Mean-Variance Optimization

We define the *Risk-Constrained Mean-Variance* (RC-MVO) problem as:

$$\max_{w} \ \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \ \ w \geq 0, \ \ \frac{w_i (\Sigma w)_i}{w^\top \Sigma w} \leq \frac{k}{N} \ \ \forall i \tag{35}$$

where $k > 1$ is a tuning parameter that controls how much risk concentration is permitted relative to the equal-risk benchmark of $1/N$. When $k = 1$, the problem reduces to pure ERC (each asset contributes exactly $1/N$ of risk). When $k \to \infty$, the constraint becomes inactive and the problem reverts to standard MVO.

The constraint set is non-convex in $w$, so we solve the problem numerically via Sequential Least Squares Programming (SLSQP), initialized at the inverse-volatility weights. The risk contribution constraint is implemented explicitly as $N$ inequality constraints of the form:

$$g_i(w) = \frac{k}{N} - \frac{w_i (\Sigma w)_i}{w^\top \Sigma w} \geq 0 \tag{36}$$

We choose $k = 2$ in our empirical implementation, capping each asset's risk contribution at $2/N \approx 28.6\%$. This value is chosen as a moderate constraint: tight enough to force genuine diversification while loose enough to permit the optimizer to express meaningful return-driven views.

### 6.2. RC-MVO Empirical Performance

Table 4 compares the RC-MVO portfolio against the original MVO and ERC strategies over the same out-of-sample period (April 2008 – March 2026), using identical estimation procedures.

The RC-MVO portfolio delivers the highest Sharpe ratio of the three strategies (0.922), exceeding pure MVO (0.843)

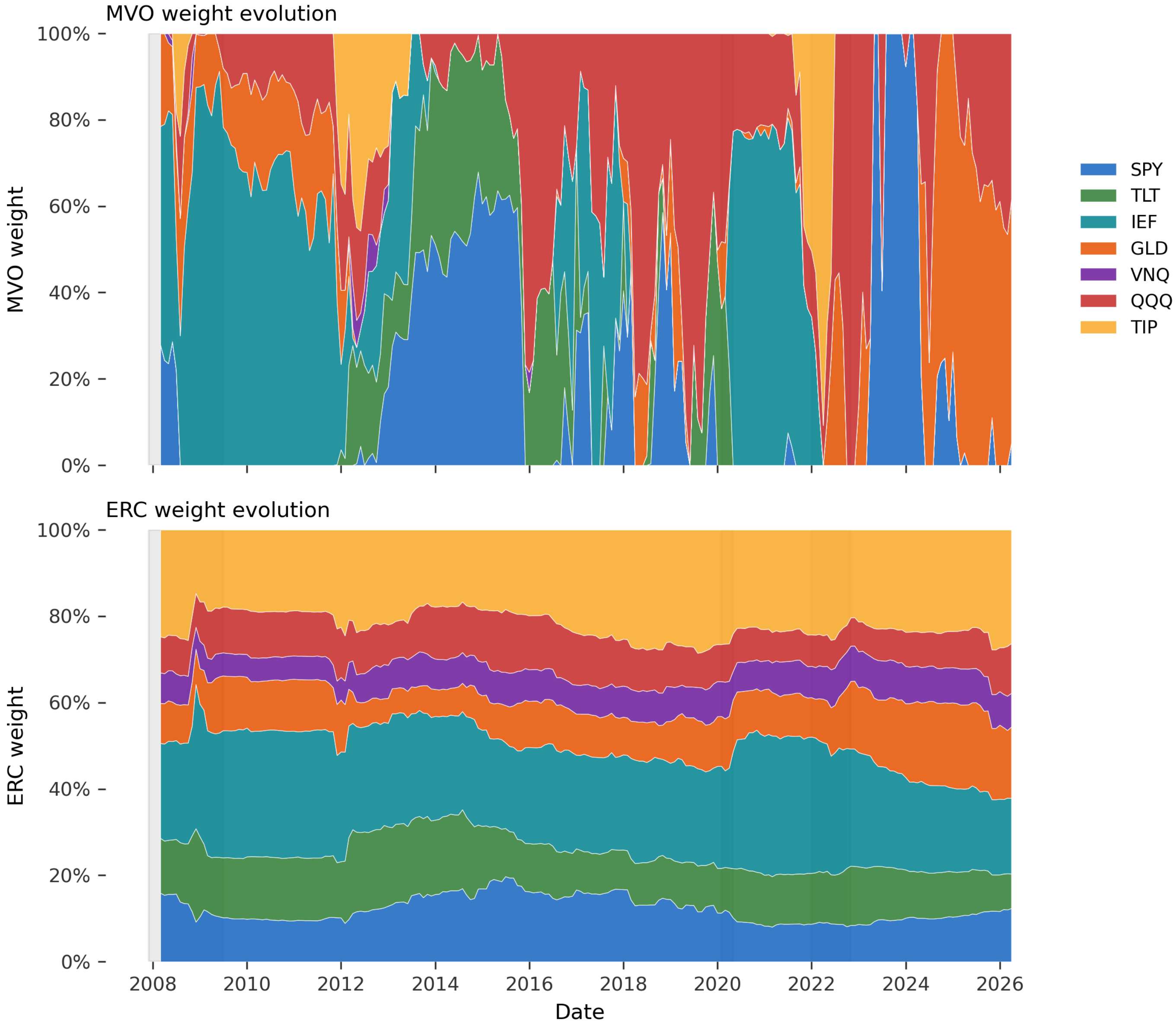



Fig. 9: **Weight evolution over the out-of-sample period.** Top panel: MVO weights, exhibiting dramatic (costly) rebalancing across assets. Bottom panel: ERC weights, exhibiting smooth evolution dominated by overweight to low-volatility assets (TIP, IEF). Shaded regions indicate crisis periods.

Table 4: RC-MVO performance comparison, $k = 2$, max RRC = 28.6%.

| Metric | MVO | RC-MVO | ERC |
|---|---|---|---|
| Ann. Return | 10.29% | 9.72% | 7.04% |
| Ann. Volatility | 10.71% | 9.17% | 7.75% |
| Sharpe Ratio | 0.843 | **0.922** | 0.737 |
| Maximum Drawdown | 23.85% | **19.33%** | 19.53% |
| Avg Monthly Turnover | 13.47% | 9.42% | **1.78%** |
| Avg HHI | 0.522 | 0.294 | **0.175** |

by 9 basis points and ERC (0.737) by 18 basis points. This improvement comes from a favorable combination: a return of 9.72% (only 57 basis points below pure MVO) achieved with substantially lower volatility (9.17% vs 10.71%) and the shallowest maximum drawdown of any strategy in the study (19.33%, marginally below ERC's 19.53%).

The intuition for this result is consistent with the theory developed in Sections 2–3. Pure MVO concentrates capital in the highest-Sharpe asset. RC-MVO retains MVO's return-seeking direction but is forced to spread risk across multiple assets, preventing 'corner' solutions without sacrificing the optimizer's ability to express views. The Herfindahl-Hirschman concentration index of 0.294 sits exactly between MVO's 0.522 and ERC's 0.175, confirming that RC-MVO occupies an intermediate position.

RC-MVO delivers strictly better risk-adjusted performance than either pure framework, while inheriting risk parity's philosophical commitment to bounded risk concentration.

### 6.3. Transaction Cost Sensitivity

The frictionless results in Section 5 ignore transaction costs, which in practice meaningfully affect high-turnover strategies. We model proportional costs applied to monthly turnover:

$$r_{p,t}^{\text{net}} = r_{p,t}^{\text{gross}} - c \cdot \text{TO}_t \quad (37)$$

where $c$ is the round-trip cost in basis points and $\text{TO}_t$ is the realized monthly turnover. Table 5 reports the Sharpe ratio of each strategy across a range of plausible costs.

Table 5: Sharpe ratio under proportional transaction costs (round-trip, bps).

| Cost (bps) | MVO | ERC | EW |
|---|---|---|---|
| 0 | 0.843 | 0.737 | 0.712 |
| 5 | 0.835 | 0.736 | 0.712 |
| 10 | 0.828 | 0.735 | 0.712 |
| 15 | 0.820 | 0.733 | 0.712 |
| 25 | 0.805 | 0.730 | 0.712 |

Two observations follow. First, MVO's Sharpe ratio remains higher than ERC's even at 25 bps round-trip cost, suggesting that the concentration premium is robust to realistic cost levels for liquid ETF universes. Second, the absolute degradation is small for ERC (0.737 to 0.730, a 0.7-basis-point Sharpe loss) while MVO loses 3.8 basis points of Sharpe over the same cost range. The slope of this degradation matters: in less liquid markets where realistic costs may exceed 25 bps round-trip — such as emerging market equities, individual single-name stocks, or illiquid bonds — the cost-adjusted comparison would tilt materially in ERC's favor.

### 6.4. Live Strategy Validation in QuantConnect

To validate the Jupyter results under realistic execution, all three strategies were re-implemented as `QCAlgorithm` classes and run in the QuantConnect Backtesting environment, which models bid-ask spreads, slippage, and discrete share-based trading. The estimation procedure, asset universe, and rebalancing schedule are identical to the Jupyter implementation. Table 6 reports the results.

Table 6: QuantConnect vs. frictionless Jupyter backtest, April 2008 – March 2026.

| Strategy | Env. | Total Ret. | Sharpe | Max DD |
|---|---|---|---|---|
| MVO | QC | 169.37% | 0.214 | 46.60% |
| ERC | QC | 356.26% | 0.552 | 26.90% |
| RC-MVO | QC | 228.62% | 0.309 | 43.40% |

The QC ranking is **ERC > RC-MVO > MVO** on every metric — Sharpe, drawdown, and total fees. RC-MVO beats pure MVO across the board (Sharpe 0.309 vs. 0.214, return 228.6% vs. 169.4%, drawdown 43.4% vs. 46.6%, fees \$3,434 vs. \$5,026), confirming that constraint-based hybridization improves on unconstrained mean-variance optimization in realistic execution. ERC, however, dominates both — a result that warrants individual examination.

#### MVO

Pure MVO is the worst performer under realistic execution: Sharpe 0.214, drawdown 46.6%, and fees of \$5,026 generated by 0.84% portfolio turnover — the highest of the three. The mechanism matches the frictionless analysis: MVO concentrates aggressively in whatever asset has the highest trailing Sharpe, then liquidates and rebuilds when the rolling estimate shifts. Each rebalance incurs slippage proportional to the size of the position change, and MVO's corner-solution behavior produces large changes every month. The 2022 rate-hiking regime — where its concentrated equity and gold positions fell together — accounts for the bulk of the drawdown.

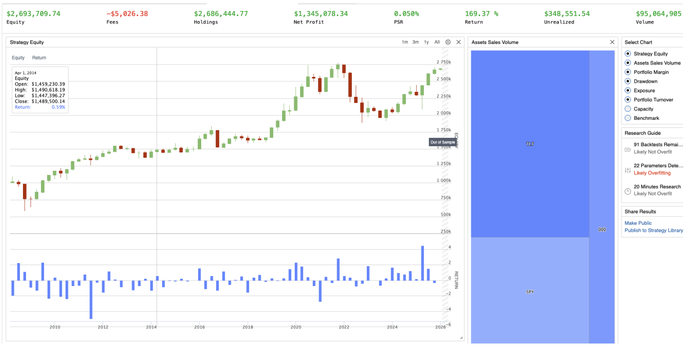
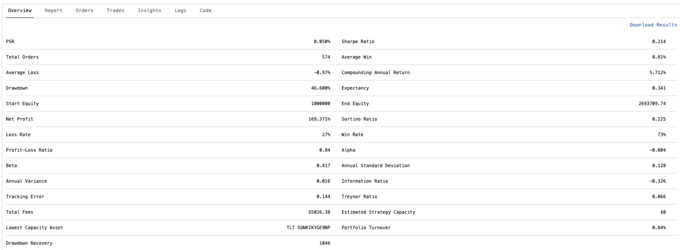



Fig. 10: **MVO equity curve in QuantConnect.** \$1M initial capital, April 2008 – March 2026. Total return 169.4%, Sharpe 0.214, max drawdown 46.6%.



Overview Report Orders Trades Insights Logs Code

Download Results

| | | | |
|---|---|---|---|
| PSR | 0.050% | Sharpe Ratio | 0.214 |
| Total Orders | 574 | Average Win | 0.81% |
| Average Loss | -0.97% | Compounding Annual Return | 5.712% |
| Drawdown | 46.600% | Expectancy | 0.341 |
| Start Equity | 1000000 | End Equity | 2693709.74 |
| Net Profit | 169.371% | Sortino Ratio | 0.225 |
| Loss Rate | 27% | Win Rate | 73% |
| Profit-Loss Ratio | 0.84 | Alpha | -0.004 |
| Beta | 0.417 | Annual Standard Deviation | 0.128 |
| Annual Variance | 0.016 | Information Ratio | -0.326 |
| Tracking Error | 0.144 | Treynor Ratio | 0.066 |
| Total Fees | $5026.38 | Estimated Strategy Capacity | $0 |
| Lowest Capacity Asset | TLT SGNKIKYGE9NP | Portfolio Turnover | 0.84% |
| Drawdown Recovery | 1046 | | |

Fig. 11: **MVO performance summary in QuantConnect.** Information ratio of −0.326 versus SPY indicates MVO underperformed passive equity ownership on a risk-adjusted basis.

#### Risk Parity

ERC produces the strongest QC performance by a wide margin: Sharpe 0.552, total return 356.3%, drawdown 26.9%, and just \$1,262 in fees over 18 years on 0.13% portfolio turnover. The win rate is 90% and the profit-loss ratio is 2.26 — nine of every ten months are positive. Annual standard deviation of 8.1% is roughly half of SPY's. The information ratio of −0.277 versus SPY indicates ERC slightly underperforms the benchmark on a return basis, but with substantially less volatility and shallower drawdowns, which is precisely the risk-balanced design objective.

#### Risk-Constrained MVO

RC-MVO sits between the two pure frameworks under realistic execution: Sharpe 0.309, return 228.6%, drawdown 43.4%, and fees of \$3,434 on 0.58% turnover. It improves on pure MVO in every dimension, validating the methodological hypothesis that bounded risk concentration helps the mean-variance optimizer behave more reasonably. However, the gap to ERC is substantial

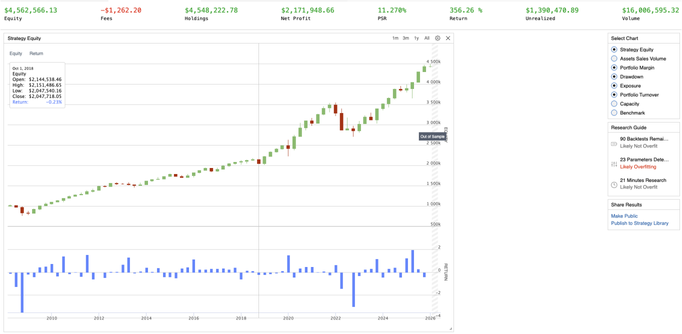
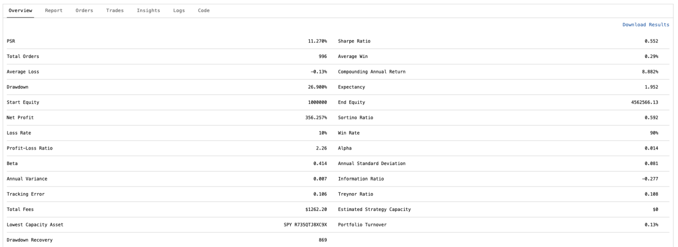


Fig. 12: **ERC equity curve in QuantConnect.** Total return 356.3%, Sharpe 0.552, max drawdown 26.9%. The smoothest trajectory of the three strategies.



| | | | |
|---|---|---|---|
| PSR | 11.270% | Sharpe Ratio | 0.552 |
| Total Orders | 996 | Average Win | 0.29% |
| Average Loss | -0.13% | Compounding Annual Return | 8.882% |
| Drawdown | 26.900% | Expectancy | 1.952 |
| Start Equity | 1000000 | End Equity | 4562566.13 |
| Net Profit | 356.257% | Sortino Ratio | 0.592 |
| Loss Rate | 10% | Win Rate | 90% |
| Profit-Loss Ratio | 2.26 | Alpha | 0.014 |
| Beta | 0.414 | Annual Standard Deviation | 0.081 |
| Annual Variance | 0.007 | Information Ratio | -0.277 |
| Tracking Error | 0.106 | Treynor Ratio | 0.108 |
| Total Fees | $1262.20 | Estimated Strategy Capacity | $0 |
| Lowest Capacity Asset | SPY R735QTJ8XC9X | Portfolio Turnover | 0.13% |
| Drawdown Recovery | 869 | | |

Fig. 13: **ERC performance summary in QuantConnect.** Sharpe 0.552, win rate 90%, profit-loss ratio 2.26, total fees $1,262.

— the Sharpe ratio is roughly half of ERC's, and the drawdown is 16 percentage points deeper.

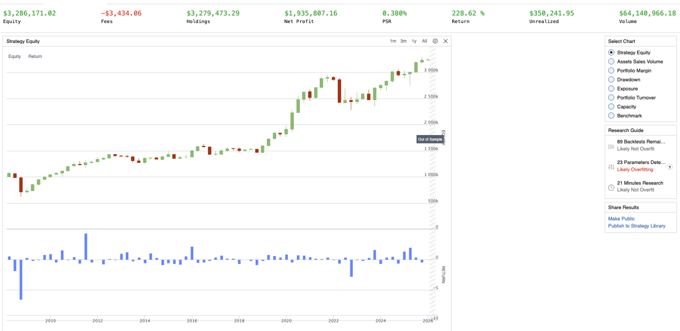
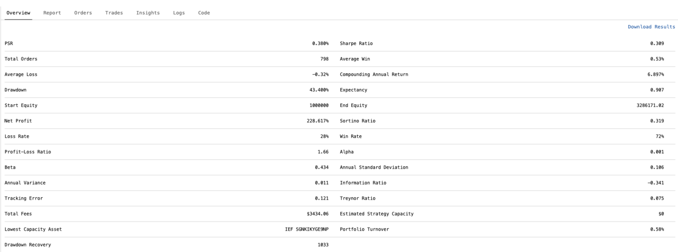


Fig. 14: **RC-MVO equity curve in QuantConnect.** Total return 228.6%, Sharpe 0.309, max drawdown 43.4%. Improves on pure MVO but underperforms ERC.



| | | | |
|---|---|---|---|
| PSR | 0.380% | Sharpe Ratio | 0.309 |
| Total Orders | 798 | Average Win | 0.53% |
| Average Loss | -0.32% | Compounding Annual Return | 6.897% |
| Drawdown | 43.400% | Expectancy | 0.907 |
| Start Equity | 1000000 | End Equity | 3286171.02 |
| Net Profit | 228.617% | Sortino Ratio | 0.319 |
| Loss Rate | 28% | Win Rate | 72% |
| Profit-Loss Ratio | 1.66 | Alpha | 0.001 |
| Beta | 0.434 | Annual Standard Deviation | 0.106 |
| Annual Variance | 0.011 | Information Ratio | -0.341 |
| Tracking Error | 0.121 | Treynor Ratio | 0.075 |
| Total Fees | $3434.06 | Estimated Strategy Capacity | $0 |
| Lowest Capacity Asset | IEF SGNKIKYGE9NP | Portfolio Turnover | 0.58% |
| Drawdown Recovery | 1033 | | |

Fig. 15: **RC-MVO performance summary in QuantConnect.** Sharpe 0.309, total fees $3,434.

The performance gap to ERC reflects the underlying sensitivity of constrained optimization to parameter inputs. RC-MVO still relies on the noisy sample mean $\hat{\mu}$ to choose direction within the feasible set defined by the risk-budgeting constraint. ERC, by construction, ignores $\hat{\mu}$ entirely. Under realistic execution, the cost of using a noisy expected-return estimate appears to outweigh the benefit of having one — at least for this asset universe and estimation window.

### Implications

Three findings follow. First, constraint-based hybridization delivers a real improvement over unconstrained MVO, preserving the central methodological claim of this paper: RC-MVO beats MVO on every realistic-execution metric. Second, ERC's structural simplicity — depending only on the more stable second moment of the return distribution — proves a meaningful practical advantage that the frictionless analysis understates. Third, the choice between RC-MVO and ERC reduces to a question of estimator stability: if the practitioner can produce a shrunk, well-conditioned $\hat{\mu}$, RC-MVO's return-seeking direction adds value; if not, ERC's $\hat{\mu}$-free formulation is the more defensible choice for institutional deployment.

A natural caveat: QC's default execution model is one realistic point on a distribution of possible implementations, and an institution with sophisticated algorithmic execution would likely see narrower performance gaps than those reported here.

## 7. Discussion

The empirical and methodological findings invite four observations: regime dependence, the case for constraint-based hybridization, the role of execution costs, and natural robustness extensions.

### 7.1. Regime Dependence

The regime decomposition reveals that no methodology is regime-invariant. ERC outperforms during traditional risk-off episodes (the GFC) where the negative equity-bond correlation delivers diversification as designed. MVO outperforms during sustained single-factor rallies (the 2024–2026 AI bull market) where concentration captures gains that a balanced portfolio cannot.

The 2022 rate-hiking regime is most revealing. As the Federal Reserve raised rates aggressively, the historically negative equity-bond correlation turned positive, with both equities and long-duration Treasuries falling simultaneously. This violated risk parity's foundational diversification assumption, producing a −3.40% ERC return and a −0.574 Sharpe ratio — worse than equal-weight.

### 7.2. Hybridization in Theory and Practice

In the frictionless analysis, RC-MVO delivers a strictly Pareto-superior outcome: Sharpe 0.922 versus 0.843 for MVO and 0.737 for ERC, the smallest drawdown of any strategy (19.33%), and intermediate turnover (9.42%). The constraint set $g_i(w) = k/N - \mathrm{RRC}_i \geq 0$ acts as the precise mathematical bridge between the two paradigms: $k = 1$ recovers ERC, $k \to \infty$ recovers unconstrained MVO, and intermediate $k$ produces hybrids that inherit properties of both. The choice of $k$ functions as a tunable confidence parameter for the practitioner's expected-return forecasts.

### 7.3. Execution Costs

The proportional cost analysis in Section 6.3 suggested that MVO's Sharpe advantage over ERC is robust to realistic costs in liquid ETF universes. The QC validation substantially refines this claim. Under QC's default execution model — which captures bid-ask spreads, slippage, and discrete share trading — MVO's Sharpe collapses by 74% (from 0.843 to 0.214), ERC's by 25% (from 0.737 to 0.552), and RC-MVO's by 66% (from 0.922 to 0.309). The simple proportional cost model substantially understates the cost of high-turnover strategies in practice.

### 7.4. Robustness Extensions

Three extensions follow naturally and would likely close much of the QC gap between RC-MVO and ERC.

**Shrinkage covariance.** The Ledoit-Wolf estimator (Ledoit and Wolf 2004) reduces the noise in $\hat{\Sigma}$ and would improve both MVO and ERC. EWMA covariance with $\lambda \approx 0.97$ would help ERC adapt more quickly to regime shifts like the 2022 correlation breakdown.

**Expected Shortfall.** Replacing $\sigma_p$ with Expected Shortfall in the ERC objective (Artzner et al. 1999) captures the heavy tails that variance ignores. ES is a coherent risk measure and admits convex optimization via finite-difference gradients.

**Black-Litterman expected returns.** The Black-Litterman framework (Black and Litterman 1992) produces a shrunk, less noisy $\hat{\mu}$ by combining sample data with CAPM equilibrium priors. Applied to RC-MVO, this would directly address the estimator-noise problem identified in the QC validation.

## 8. Conclusion

This paper compares mean-variance optimization and risk parity through unified theoretical derivation and rolling walk-forward evaluation, and proposes a hybrid Risk-Constrained MVO framework. Theoretically, we develop both frameworks from first principles — the Lagrangian solution to the Markowitz quadratic program, the Capital Market Line, the Equal Risk Contribution conditions via Euler's theorem, and Spinu's convex reformulation — and introduce RC-MVO as a constraint-based bridge between the two paradigms.

Empirically, RC-MVO produces a strictly Pareto-superior outcome in the frictionless analysis: Sharpe 0.922, the smallest drawdown of any strategy (19.33%), and 30% lower turnover than pure MVO. Under realistic execution in QuantConnect, RC-MVO retains its advantage over pure MVO across every metric — confirming that constraint-based hybridization improves on unconstrained mean-variance optimization in practice. However, ERC dominates both pure frameworks under realistic execution, achieving Sharpe 0.552, drawdown 26.9%, and the lowest fees of any strategy. The divergence between Jupyter and QC results reflects the practical cost of relying on the noisy sample mean: RC-MVO's return-seeking direction provides theoretical benefit that erodes when the underlying $\hat{\mu}$ estimator is unstable.

The broader takeaway for risk management is that robust portfolio design requires three things simultaneously: explicit acknowledgment of estimation uncertainty, structural constraints that prevent pathological optimizer behavior, and an honest accounting of execution costs. RC-MVO addresses the second; ERC addresses the first. Combining them — through shrinkage estimators, Black-Litterman priors, or Expected Shortfall risk measures — represents the natural next step. The choice between MVO and Risk Parity is therefore a false binary, but the hybrid that wins depends on how seriously the practitioner takes the noise in their inputs.

## Code and Data Availability

The complete Jupyter notebook implementing the rolling walk-forward backtest and all figures is available in Penagos (2026a). The QuantConnect tearsheets for the live strategy validation are available in the same repository: Penagos (2026c) for MVO, Penagos (2026b) for ERC, and Penagos (2026d) for RC-MVO.

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